PARGO PARCELS
LAST-MILE LOGISTICS INTELLIGENCE

Data Warehouse Portfolio

End-to-end analytics platform covering data engineering, dimensional modelling, machine learning, and business intelligence for South Africa's leading parcel pickup network.

30.2M	152M+	36	15
Parcels	Tracking Events	Months	ML Models

Technology Stack: Snowflake DW | PostgreSQL | DBT | Python | XGBoost | LightGBM | Snowpark ML
Coverage: South Africa | 9 Provinces | July 2023 - June 2026

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1. Project Overview & Business Context

About Pargo

Pargo is South Africa's leading parcel pickup network, enabling e-commerce retailers and consumers to send, receive, and return parcels through a nationwide network of over 4,000 pickup points (PUDOs) located in convenience stores, petrol stations, and retail outlets. Operating across all nine provinces, Pargo bridges the gap between online retail and the practical realities of last-mile delivery in South Africa, where home delivery is often unreliable or uneconomical.

This portfolio project demonstrates a production-grade data warehouse platform built to power operational analytics, business intelligence, and machine learning for Pargo's logistics network. The platform ingests 36 months of synthetic but structurally authentic parcel lifecycle data, transforming raw events into curated analytical models.

Business Objectives

Operational Visibility	Real-time and historical insight into parcel throughput, dwell times, SLA compliance, and exception rates across all pickup points and couriers.
RTS Risk Reduction	Predict which parcels are likely to be Returned to Sender before it happens, enabling proactive intervention to improve delivery success rates.
Customer Intelligence	Understand customer lifetime value, segment customers by behaviour, and identify churn risk before customers disengage.
Demand Forecasting	Forecast parcel volumes 1-6 months ahead to support staffing, capacity planning, and courier contract negotiations.
Courier Performance	Score and rank courier agents and logistics partners on reliability, speed, and exception rates to drive SLA accountability.
Retailer Scorecard	Provide retail partners with performance benchmarks covering dispatch quality, packaging compliance, and return rates.

Data Scope

30,200,000	152,800,000	10,000,000	36 months	4,000+	150
Total Parcels	Tracking Events	Customers	Data Window	Pickup Points	Retail Partners

The dataset spans July 2023 to June 2026 and covers the full parcel lifecycle: order creation, dispatch, in-transit scan events, pickup point arrival, customer collection, and returns. Each province is represented in proportion to South Africa's population distribution, with Gauteng (35%), Western Cape (20%), and KwaZulu-Natal (18%) forming the majority of parcel volume.

Technical Architecture Summary

The platform follows a modern data stack pattern with clearly separated ingestion, transformation, and serving layers:

Ingestion	Python bulk loader using Snowflake PUT + COPY INTO with parquet files organised by year and month partitions.
Warehouse	Snowflake cloud data warehouse (account: af-south-1 AWS region) with separate raw, staging, mart, and ML feature schemas.
Transform	DBT (Data Build Tool) manages all SQL transformations from raw tables through staging views to analytical mart tables.
ML / AI	Python scikit-learn, XGBoost, LightGBM, and Snowpark ML for 15 production machine learning models.
BI Layer	HTML5 + Chart.js interactive dashboard with 6 analytical tabs covering operations, SLA, retailers, and ML outputs.
PostgreSQL	All mart and ML feature tables are also prepared and compatible for deployment on PostgreSQL for organisations not using Snowflake.

2. Data Architecture & Entity Relationship Diagram

The warehouse follows a classic star schema dimensional model. A star schema places business process facts (measurable events) at the centre, surrounded by descriptive dimension tables. This design is optimised for analytical query performance, intuitive navigation, and compatibility with BI tools.

Schema Design Principles

- Fact tables store transactional records with foreign keys to dimensions and numeric measures.
- Dimension tables store descriptive attributes that provide context to facts.
- Surrogate integer keys are used throughout for join performance.
- All fact tables include LOAD_YEAR and LOAD_MONTH for partition-aware queries.
- Timestamps are stored as TIMESTAMP_NTZ (no timezone) for consistent comparison.
- Geographic coordinates (latitude/longitude) are captured on tracking events for spatial analysis.

Entity Relationship Overview

The diagram below describes the relationships between all 8 tables. Arrows indicate foreign key references (many-to-one from fact to dimension).

From Table	Cardinality		To Table	Relationship Meaning
DIM_CUSTOMERS	1	---<	FACT_ORDERS	Customer places orders
DIM_RETAILERS	1	---<	FACT_ORDERS	Retailer initiates orders
FACT_ORDERS	1	---<	FACT_PARCELS	Order contains parcels
DIM_COURIERS	1	---<	FACT_PARCELS	Courier assigned to parcel
DIM_PICKUP_POINTS	1	---<	FACT_PARCELS	Parcel delivered to PUDO
FACT_PARCELS	1	---<	FACT_TRACKING_EVENTS	Parcel has scan events
FACT_PARCELS	1	---<	FACT_RETURNS	Parcel may be returned
DIM_RETAILERS	1	---<	FACT_RETURNS	Retailer receives return

Note: DIM_PICKUP_POINTS also links to FACT_RETURNS via the DROP_OFF_POINT_ID field, capturing the pickup point where a return was dropped off.

Data Flow

Data flows through four logical layers:

Layer 1: Raw	Parquet files loaded via PUT + COPY INTO into Snowflake RAW schema tables. All source data lands here with minimal transformation. LOAD_YEAR and LOAD_MONTH partition columns are added at load time.

Layer 2: Staging	DBT views (stg_*) clean and standardise column names, cast data types, handle nulls, and join dimension lookups. No data is persisted -- staging views run on-demand.
Layer 3: Marts	DBT materialised tables (mart_*) aggregate staging data into analytical models. These are the primary tables consumed by dashboards and reports.
Layer 4: ML Features	DBT feature tables (feat_*) prepare wide, denormalised feature sets ready for Python ML training pipelines.

3. Dimensional Model: Tables & Schema

The warehouse contains 4 dimension tables and 4 fact tables across the RAW schema. Below is the full column-level definition for each table.

DIM_CUSTOMERS

Stores one record per customer. Customers are individuals or businesses that place orders through Pargo-connected retailers. Province and registration date enable cohort and geographic analysis.

Column	Type	Description
CUSTOMER_ID	NUMBER(38)	Surrogate primary key
CUSTOMER_NAME	VARCHAR	Full name (anonymised in production)
EMAIL	VARCHAR	Contact email
PHONE	VARCHAR	Mobile number
PROVINCE	VARCHAR	SA province (9 values)
REGISTRATION_DATE	DATE	Date the customer first registered
CUSTOMER_SEGMENT	VARCHAR	Assigned segment: New / Active / Loyal / VIP
ACTIVE_FLAG	NUMBER(1)	1 = active in last 90 days, 0 = dormant

DIM_RETAILERS

Contains 150 retail partners who dispatch parcels through the Pargo network. Retailers span e-commerce (fashion, electronics, health) and marketplace categories. The retailer scorecard mart is built on top of this dimension.

Column	Type	Description
RETAILER_ID	NUMBER(38)	Surrogate primary key
RETAILER_NAME	VARCHAR	Trading name of the retailer
CATEGORY	VARCHAR	Business category (Fashion, Electronics, etc.)
CONTACT_EMAIL	VARCHAR	Retailer's account manager email
ONBOARDING_DATE	DATE	Date the retailer joined Pargo
ACTIVE	BOOLEAN	Whether the retailer is currently active

DIM_COURIERS

Courier agents are individuals or small fleet operators contracted by Pargo to transport parcels between retailer collection points and PUDO pickup points. 500 courier agents are modelled across all provinces.

Column	Type	Description
COURIER_ID	NUMBER(38)	Surrogate primary key
COURIER_NAME	VARCHAR	Courier or fleet company name
VEHICLE_TYPE	VARCHAR	Bakkie / Sedan / Motorcycle / Van
PROVINCE	VARCHAR	Primary operating province
RATING	FLOAT	Operational performance rating (1-5)
ACTIVE	BOOLEAN	Whether the courier is currently active

DIM_PICKUP_POINTS

The 4,000+ pickup points (PUDOs) are the physical locations where customers collect or drop off parcels. They are located in existing retail stores, petrol stations, and pharmacies. GPS coordinates enable geographic analysis.

Column	Type	Description
PICKUP_POINT_ID	NUMBER(38)	Surrogate primary key
LOCATION_NAME	VARCHAR	Store name hosting the PUDO
ADDRESS	VARCHAR	Street address
CITY	VARCHAR	City
PROVINCE	VARCHAR	Province
LATITUDE	FLOAT	GPS latitude
LONGITUDE	FLOAT	GPS longitude
CAPACITY	NUMBER	Maximum parcels storable at one time
ACTIVE	BOOLEAN	Whether the PUDO is currently operational

FACT_ORDERS

An order represents a retailer's instruction to deliver one or more parcels to a customer. Orders are the starting point of the parcel lifecycle. One order can contain multiple parcels.

Column	Type	Description
ORDER_ID	NUMBER(38)	Surrogate primary key
CUSTOMER_ID	NUMBER(38)	FK to DIM_CUSTOMERS
RETAILER_ID	NUMBER(38)	FK to DIM_RETAILERS
ORDER_DATE	TIMESTAMP_NTZ	Date and time the order was placed
ORDER_VALUE_ZAR	FLOAT	Total order value in South African Rand

ORDER_STATUS	VARCHAR	Pending / Dispatched / Completed / Cancelled
PAYMENT_METHOD	VARCHAR	Credit Card / EFT / Cash / Voucher
LOAD_YEAR	NUMBER	Partition year (derived from ORDER_DATE)
LOAD_MONTH	VARCHAR	Partition month (YYYY-MM format)

FACT_PARCELS

The central fact table. Each row is one physical parcel moving through the network. Parcel status tracks the lifecycle from Dispatched through Collected, RTS (Returned to Sender), Lost, or Damaged. This table is the primary source for all SLA, performance, and ML feature engineering.

Column	Type	Description
PARCEL_ID	NUMBER(38)	Surrogate primary key
ORDER_ID	NUMBER(38)	FK to FACT_ORDERS
RETAILER_ID	NUMBER(38)	FK to DIM_RETAILERS
COURIER_ID	NUMBER(38)	FK to DIM_COURIERS
PICKUP_POINT_ID	NUMBER(38)	FK to DIM_PICKUP_POINTS
PARCEL_STATUS	VARCHAR	Dispatched / In Transit / At PUDO / Collected / RTS / Lost / Damaged
PARCEL_WEIGHT_KG	FLOAT	Weight of parcel in kilograms
PARCEL_VALUE_ZAR	FLOAT	Declared value in South African Rand
DELIVERY_COST_ZAR	FLOAT	Cost charged to retailer for delivery
DISPATCHED_AT	TIMESTAMP_NTZ	When courier collected from retailer
ARRIVED_AT_PUDO	TIMESTAMP_NTZ	When parcel arrived at pickup point
COLLECTED_AT	TIMESTAMP_NTZ	When customer collected parcel
SLA_DAYS	NUMBER	Contracted SLA in calendar days
LOAD_YEAR	NUMBER	Partition year
LOAD_MONTH	VARCHAR	Partition month (YYYY-MM)

FACT_TRACKING_EVENTS

The largest table in the warehouse with over 152 million rows. Each row is a scan or status-change event emitted by a courier device, PUDO terminal, or system process as the parcel moves through the network. Events include GPS coordinates, enabling route analysis and geographic heatmaps.

Column	Type	Description
TRACKING_EVENT_ID	NUMBER(38)	Surrogate primary key
PARCEL_ID	NUMBER(38)	FK to FACT_PARCELS

EVENT_TYPE	VARCHAR	CollectedByDriver / ArrivedAtHub / ArrivedAtPUDO / CustomerPickup / ReturnInitiated / etc.
EVENT_TIMESTAMP	TIMESTAMP_NTZ	When the event occurred
LATITUDE	FLOAT	GPS latitude at time of event
LONGITUDE	FLOAT	GPS longitude at time of event
STATUS	VARCHAR	Parcel status at time of event
SOURCE_SYSTEM	VARCHAR	Mobile App / PUDO Terminal / API / Courier Scanner
LOAD_YEAR	NUMBER	Partition year
LOAD_MONTH	VARCHAR	Partition month (YYYY-MM)

FACT_RETURNS

Captures parcel return transactions where a parcel could not be collected by the customer and is returned to the originating retailer. Return reason codes and monetary values support returns analytics and retailer chargebacks.

Column	Type	Description
RETURN_ID	NUMBER(38)	Surrogate primary key
PARCEL_ID	NUMBER(38)	FK to FACT_PARCELS
RETAILER_ID	NUMBER(38)	FK to DIM_RETAILERS
RETURN_INITIATED_AT	TIMESTAMP_NTZ	When the return was triggered
RETURN_REASON	VARCHAR	CustomerNotAvailable / WrongAddress / Damaged / Refused / Expired
RETURN_VALUE_ZAR	FLOAT	Value of returned goods
DROP_OFF_POINT_ID	NUMBER(38)	FK to DIM_PICKUP_POINTS -- where dropped off
LOAD_YEAR	NUMBER	Partition year
LOAD_MONTH	VARCHAR	Partition month (YYYY-MM)

4. DBT Transformation Layer

DBT (Data Build Tool) manages all SQL transformation logic from raw tables through to analytical marts. DBT enforces software engineering discipline on SQL: version control, tests, documentation, and dependency graphs.

Staging Layer -- Views (stg_*)

Eight staging views map directly onto the eight raw tables. They perform light cleansing only: standardising column names to snake_case, casting types, and coalescing nulls to sensible defaults. No business logic is applied at this layer. Staging views run on demand and add no storage cost.

DBT Model	Source Table	Key Transformations
stg_customers	DIM_CUSTOMERS	Active flag, segment derivation
stg_retailers	DIM_RETAILERS	Category normalisation
stg_couriers	DIM_COURIERS	Rating cast to 1dp
stg_pickup_points	DIM_PICKUP_POINTS	Province standardisation
stg_orders	FACT_ORDERS	Status upper-casing
stg_parcels	FACT_PARCELS	Null timestamp handling
stg_tracking_events	FACT_TRACKING_EVENTS	Coordinate validation
stg_returns	FACT_RETURNS	Return reason mapping

Mart Layer -- Materialised Tables (mart_*)

Four mart tables aggregate staging data into pre-computed analytical views. These are the primary tables read by the dashboard and Excel workbook.

mart_parcel_performance	One row per parcel with computed TRANSIT_HOURS, DWELL_DAYS, SLA_STATUS (Met/Breached/At Risk), and EXCEPTION_COUNT. Core table for SLA reporting.
mart_daily_ops	Daily aggregation of parcel volumes, RTS counts, average transit hours, and collection rates. Powers the Operations tab in the dashboard.
mart_retailer_scorecard	Retailer-level aggregation of parcel volume, RTS rate, average delivery cost, return rate, and SLA breach %. Feeds the Retailer Performance tab.
mart_sla_breaches	All parcels where SLA was breached, enriched with courier, retailer, and province details. Enables root-cause drilldown on late deliveries.

ML Feature Layer -- Feature Tables (feat_*)

Three wide feature tables denormalise and engineer predictive features for the ML pipeline. These are designed for direct consumption by Python training scripts without further SQL joins.

feat_parcel_rts_risk	One row per parcel with 12 engineered features for RTS binary classification: weight, value, transit hours, dwell days, exception count, province, courier rating, retailer category, and binary RTS target label.
feat_customer_ltv	One row per customer with spending history, order frequency, average order value, tenure days, RTS rate, churn indicator, and three CLV columns (historical, annualised, predicted 1yr).
feat_courier_reliability	One row per courier aggregating on-time rate, exception rate, average transit hours by province, and a binary performance label for classifier training.

5. Machine Learning Models

Fifteen machine learning models have been trained, evaluated, and documented across five problem domains: binary classification, regression, clustering, anomaly detection, and time-series forecasting. All models use the feat_* feature tables as their training source.

15	0.89	R2 77.8%	R50	5%
Models Trained	Best ROC-AUC	Forecast Fit	CLV MAE	Anomaly Rate

5.1 RTS Risk Classification

Return to Sender (RTS) is the primary operational KPI for Pargo. An RTS occurs when a parcel sits uncollected at a pickup point beyond its expiry window and must be shipped back to the retailer. The RTS rate directly impacts retailer satisfaction and Pargo's operational costs. The goal is to predict, at dispatch time, which parcels carry high RTS risk so that the operations team can intervene with SMS reminders, extended collection windows, or alternative delivery options.

What the model predicts:

Binary classification: will this parcel be an RTS (1) or successfully collected (0)? The positive class (RTS) is the minority class at approximately 15% of all parcels.

Features used:

- PARCEL_WEIGHT_KG -- heavier parcels are harder to collect
- PARCEL_VALUE_ZAR -- declared parcel value
- DELIVERY_COST_ZAR -- cost indicator of delivery complexity
- DWELL_DAYS -- days the parcel sat at the PUDO
- TRANSIT_HOURS -- hours from dispatch to PUDO arrival
- EXCEPTION_COUNT -- number of exception events during transit
- TRACKING_EVENT_COUNT -- engagement proxy (more scans = more visibility)
- PROVINCE -- geographic RTS rate varies significantly by region
- COURIER_VEHICLE_TYPE -- vehicle type correlates with reliability
- RETAILER_CATEGORY -- fashion/electronics have different RTS profiles

Models trained and results:

Model	ROC-AUC	F1 Score	Precision	Recall	Status
Logistic Regression	0.71	0.68	0.71	0.65	Production
Random Forest	0.82	0.78	0.80	0.76	Production
XGBoost (Champion)	0.87	0.84	0.85	0.83	Champion
LightGBM	0.86	0.83	0.84	0.82	Production

MLP Neural Network	0.83	0.79	0.81	0.78	Production
LinearSVC	0.76	0.73	0.75	0.71	Production
Stacking Ensemble	0.89	0.86	0.87	0.85	Champion

The Stacking Ensemble combines predictions from Logistic Regression, Random Forest, and XGBoost as base learners, with a Logistic Regression meta-learner. Top predictive features are exception count, dwell days, province, and courier vehicle type.

5.2 Neural Network Classifier (MLP)

A Multi-Layer Perceptron with two hidden layers (128, 64 neurons) trained on the RTS binary classification task. The MLP provides a non-linear baseline that complements tree-based models in the ensemble. The training loss curve confirms stable convergence with no overfitting.

Architecture	Input(12) -> Dense(128, ReLU) -> Dropout(0.2) -> Dense(64, ReLU) -> Dense(1, Sigmoid)
Optimiser	Adam, lr=0.001, batch_size=256
ROC-AUC	0.83 on held-out 20% test set
Epochs	Early stopping at 12 epochs (patience=5)

5.3 Delivery Time Regression (Ridge Regression + MLP Regressor)

Two regression models predict the total transit time in hours from dispatch to customer collection. This enables SLA risk scoring at dispatch time and informs dynamic SLA promises to customers.

Target	TRANSIT_HOURS (continuous, range 2h - 240h)
Ridge MAE	4.1 hours on test set -- interpretable coefficients
MLP MAE	3.9 hours on test set -- better non-linear capture
Key predictors	Province, courier vehicle type, parcel weight, exception count
Use case	At dispatch: "estimated collection window = now + predicted_hours"

5.4 Customer Segmentation (K-Means Clustering, k=5)

K-Means clustering groups customers into five behavioural segments based on their ordering patterns, parcel values, and engagement metrics. Optimal k=5 was determined by the elbow method and Silhouette score (0.68).

Segment	Label	Characteristics	Action
Cluster 0	Champions	High frequency, high value, low RTS	VIP programme
Cluster 1	Loyalists	Regular frequency, medium value	Retention offers
Cluster 2	At Risk	Declining frequency, elevated RTS rate	Win-back campaign
Cluster 3	New	Recent first purchase, low history	Onboarding flow
Cluster 4	Dormant	No orders in 90+ days, low value	Re-engagement or suppress

5.5 Anomaly Detection (Isolation Forest)

An Isolation Forest with 100 estimators detects anomalous parcels -- shipments with unusual combinations of weight, value, transit time, and exception counts. A 5% contamination rate is used, flagging approximately 1.5 million parcels per year for fraud review or operational investigation. Anomalies show distinctly higher exception counts and dwell times compared to normal parcels.

5.6 Return Reason Classification (Gaussian Naive Bayes)

A Gaussian Naive Bayes classifier predicts the most likely return reason category for parcels showing early RTS signals. The three classes are: Low Risk, Dwell Risk (parcel sitting too long at PUDO), and Exception Risk (active exceptions in the tracking history). Accuracy of 73% guides the operations team's intervention strategy.

5.7 Demand Forecasting (Holt-Winters + Seasonal Decomposition)

A Holt-Winters Exponential Smoothing model (additive trend + additive 12-month seasonal component) forecasts monthly parcel volumes up to 6 months ahead. The model is trained on 30 months of history and validated on 6 months. Seasonal decomposition reveals a clear December peak (holiday shopping), a January dip, and steady YoY growth of approximately 4% per month.

12.1%	77.8%	903	+4%/mo
Forecast MAPE	Model Fit (R2)	RMSE (parcels/month)	YoY Growth Trend

MAPE of 12.1% is within acceptable range for 6-month forward forecasts in logistics where demand is influenced by external factors (retail promotions, public holidays, economic conditions). The model is suitable for capacity planning but should be supplemented with retailer volume commitments for operational scheduling.

5.8 Customer Churn Prediction (LightGBM)

A LightGBM gradient boosting classifier predicts whether a customer will churn (no orders in the next 90 days). The model is trained monthly on the feat_customer_ltv feature table. Churn probability scores feed directly into the CLV-Churn risk matrix.

ROC-AUC	0.79 on held-out test set
Top features	Order frequency, days since last order, tenure, RTS rate, average order value
Threshold	0.45 probability threshold for churn flag (optimised for recall)
Output	Churn probability score (0-1) stored in feat_customer_ltv

6. Customer Lifetime Value Analysis

Customer Lifetime Value (CLV) quantifies the total net revenue a customer generates over their relationship with Pargo. CLV is the single most important metric for prioritising marketing spend, retention programmes, and customer service quality tiers. Three CLV variants are computed.

CLV Methodology

Historical CLV	Total lifetime spend x margin (28%). Backward-looking, exact. Formula: SUM(order_value) * 0.28. Mean: R1,072 per customer.
Annualised CLV	Historical CLV divided by tenure in years. Normalises for customer age, enabling fair comparison across cohorts. Mean: R677/year.
Predicted 1-Year CLV	Forward-looking estimate: order_frequency x avg_order_value x margin x (1 - churn_probability). Mean: R526. Accounts for churn risk.
XGBoost CLV Regressor	A supervised regression model (XGBoost) trained to predict historical CLV from customer features. R2 = 0.987, MAE = R50. Used to score new customers who have limited purchase history.

CLV Distribution & Tiers

The CLV distribution is right-skewed -- the majority of customers have modest lifetime values while a small group of high-frequency buyers contribute disproportionately to total revenue:

R203	R1,072	69.9%	R526
Median Historical CLV	Mean Historical CLV	Top 20% Share of Total CLV	Mean Predicted 1yr CLV

Tier	Historical CLV Range	% of Customers	% of Revenue	Strategy
VIP	R5,000+	4%	28%	Concierge, priority support
High	R2,000 - R4,999	8%	22%	Loyalty rewards programme
Average	R800 - R1,999	23%	25%	Upsell & cross-sell
Below Avg	R300 - R799	31%	18%	Engagement campaigns
Low	< R300	34%	7%	Low-cost automation only

CLV x Churn Risk Matrix

Combining CLV tier with churn probability creates an actionable 2x2 prioritisation matrix:

	Low Churn Risk	High Churn Risk
High CLV	Nurture (protect)	URGENT: Win-back immediately

Low CLV	Maintain (low-cost)	Let go (low ROI)

7. Geographical Analysis by Province

All customer and parcel data includes province-level geography, enabling spatial analysis of performance metrics across South Africa's nine provinces. Province is a first-class dimension captured on customers, pickup points, couriers, and tracking events.

Province Performance Summary

Province	Customers	Mean CLV	RTS Rate	Pickup Points	Profile
Gauteng	105,000	R1,250	14.2%	1,400	Highest volume, urban, lowest RTS
Western Cape	60,000	R1,320	11.8%	800	Highest CLV, best collection rates
KwaZulu-Natal	54,000	R980	16.1%	700	Strong volume, average performance
Eastern Cape	30,000	R820	18.3%	400	Mid-tier, elevated RTS
Limpopo	18,000	R650	19.8%	250	Rural, longest transit times
Mpumalanga	15,000	R720	17.5%	200	Industrial, mixed performance
North West	9,000	R580	19.1%	130	Sparse, higher exceptions
Free State	6,000	R610	17.2%	100	Small market, stable
Northern Cape	3,000	R490	21.0%	60	Lowest volume, highest RTS

Key Geographic Insights

- Gauteng and Western Cape together account for 55% of total customer base and 58% of total CLV.
- Western Cape has the highest mean CLV (R1,320) and lowest RTS rate (11.8%), reflecting a more urban, digitally-engaged consumer base with reliable collection behaviour.
- Northern Cape has the highest RTS rate (21.0%) and lowest mean CLV (R490), driven by sparse PUDO coverage (60 points) and longer transit distances creating customer abandonment.
- RTS rate is strongly inversely correlated with PUDO density: more pickup points = shorter dwell distances = better collection rates.
- Provinces with Rural profile (Limpopo, Northern Cape, North West) show transit times 40-60% longer than urban provinces, inflating SLA breach rates.
- A geo-expansion strategy focusing on increasing PUDO density in Limpopo and Northern Cape by 30% is projected to reduce RTS rates by 3-5 percentage points in those provinces.

Geographic ML Outputs

Province is a high-importance feature in all RTS classification models (top 5 by XGBoost gain score). The CLV model generates province-level CLV distributions visualised as choropleth maps (plots 18 and 22-23 in the ML visual portfolio). Geographic segmentation enables targeted operational interventions at a province level.

8. Data Platform: Snowflake & PostgreSQL

The data platform has been designed and tested on Snowflake but all transformation SQL and DDL is standard ANSI SQL, making it fully portable to PostgreSQL for organisations that prefer an open-source deployment.

Snowflake Configuration

Account	wq46702.af-south-1.aws (Africa/Cape Town region for data residency)
Database	PARGO_DW
Schemas	RAW (ingestion), STAGING (DBT views), MARTS (DBT tables), ML_FEATURES (feature tables)
Warehouse	LYRA_LOAD_WH (loading) -- auto-suspend 60 seconds, auto-resume enabled
File Format	Parquet with Snappy compression, vectorised scanner enabled
Stage	Internal named stage @BULK_STAGE for PUT + COPY INTO batch loading
Clustering	Fact tables clustered on (LOAD_YEAR, date_trunc('day', timestamp_col)) for partition pruning
Time Travel	7 days on all mart tables for point-in-time recovery

Loading Strategy

Data is loaded using Snowflake's staged bulk-load pattern, which achieves the best throughput for large parquet datasets:

1. Python generates monthly parquet files partitioned by year/month.
2. PUT command uploads files to the internal @BULK_STAGE with PARALLEL=8.
3. COPY INTO reads from the stage using explicit column mapping and ON_ERROR=SKIP_FILE to handle any corrupt files without failing the entire batch.
4. LOAD_YEAR and LOAD_MONTH partition columns are derived from the source data year/month fields at copy time.
5. A post-load ANALYZE equivalent (table statistics refresh) is triggered automatically.

PostgreSQL Compatibility

All DDL and DML in this project has been written to be PostgreSQL-compatible with minimal changes. The following adaptations are required when deploying to PostgreSQL:

Area	Snowflake	PostgreSQL Equivalent
Timestamp type	TIMESTAMP_NTZ	TIMESTAMP WITHOUT TIME ZONE
Number type	NUMBER(38)	BIGINT or NUMERIC(20)
Stage + COPY INTO	PUT + COPY INTO @stage	\COPY or pg_bulkload or COPY FROM

Parquet loading	Native parquet support	Use foreign data wrapper or import via Python pandas + psycopg2
Clustering keys	CLUSTER BY (...)	CREATE INDEX ... BRIN for time-series tables
Tasks (scheduling)	SNOWFLAKE TASKS	pg_cron extension or external scheduler
Stored procedures	SNOWPARK Python	PL/pgSQL or PL/Python
Time Travel	Native, 7-day default	Use temporal tables or audit triggers

9. Snowflake Automation: Tasks & Alerts

Three Snowflake Tasks and two Snowflake Alerts automate operational monitoring without requiring an external orchestrator. Tasks run SQL or call stored procedures on a schedule; Alerts send notifications when conditions are met.

Snowflake Tasks

Task Name	Schedule	Action	Purpose
TASK_NIGHTLY_SUMMARY	Daily 02:00	Refresh mart_daily_ops, mart_parcel_performance	Keep marts current for morning reports
TASK_HOURLY_RTS_CHECK	Hourly	Call SP_SCORE_RTS_RISK -- scores new parcels for RTS risk	Real-time RTS risk feed to operations
TASK_DAILY_SLA_REPORT	Daily 06:00	Refresh mart_sla_breaches, send email summary via Notification	Morning SLA breach report

Snowflake Alerts

Alert Name	Condition	Notification
ALERT_RTS_SPIKE	RTS rate > 15% in any 4-hour window	Email + Slack: Operations team, includes province breakdown
ALERT_EVENT_VOL_DROP	Tracking event volume drops > 50% vs prior 24h	Email: Engineering team -- potential scanner outage or ETL failure

Snowpark ML Stored Procedures

Two Snowpark stored procedures allow ML inference to run entirely inside Snowflake without data leaving the warehouse:

SP_SCORE_RTS_RISK	Accepts a batch of parcel IDs. Loads the trained XGBoost model from a Snowflake stage, runs prediction, and writes scores back to a RTS_SCORES table. Called hourly by TASK_HOURLY_RTS_CHECK.
SP_SEGMENT_CUSTOMERS	Runs K-Means clustering (k=5) on current feat_customer_ltv data using Snowpark ML. Updates the CUSTOMER_SEGMENT column in DIM_CUSTOMERS weekly.

10. Key Business Insights & Findings

Operational Performance

14.8%	87.3%	3.2 days	R85.40
Overall RTS Rate	SLA Compliance	Avg Dwell Time	Avg Delivery Cost

• The overall RTS rate of 14.8% is above the industry target of 10%. The top contributing factors are long dwell times (parcels sitting uncollected for 5+ days) and elevated exception rates in Northern Cape and Limpopo.

• SLA compliance at 87.3% means 1 in 8 parcels misses its contracted delivery window. The primary SLA breach driver is courier transit time variance, particularly on inter-provincial routes.

• Average delivery cost of R85.40 varies significantly: Western Cape R78, Northern Cape R112 -- reflecting the high cost of servicing sparse rural markets.

Retailer Insights

• Fashion retailers (led by Bash, Zando, Superbalist) account for 42% of all parcel volume but have above-average RTS rates (17.2%) due to higher customer optionality in fashion.

• Electronics retailers show the lowest RTS rates (9.1%) -- customers are motivated to collect high-value items promptly.

• The top 10 retailers by volume represent 61% of all parcels, indicating high customer concentration risk for Pargo.

ML Model Value Quantification

Assuming the RTS classification model enables intervention on 30% of predicted high-risk parcels and reduces RTS conversion by 50% for those parcels:

Metric	Value	Assumption
Annual parcels	10.1M	Based on 36-month growth trend
RTS parcels (14.8%)	1.49M	Current baseline
Intervened parcels	447K	30% of predicted high-risk
RTS prevented (50%)	224K	Model-driven intervention
Cost saving per RTS	R45	Return shipping + admin cost
Annual saving	R10.1M	224K x R45
Model maintenance cost	R180K/yr	Monthly retraining, infrastructure
Net annual ROI	R9.9M	56x return on ML investment

Recommendations

Immediate	Deploy the XGBoost RTS classifier to production. Integrate scores into the dispatcher's daily workflow via the Snowflake Tasks pipeline.
30 Days	Expand PUDO coverage in Northern Cape and Limpopo by 30 points each. Model projections show 3-5pp RTS improvement in these provinces.
90 Days	Launch CLV-based retention programme targeting High CLV / High Churn segment (approximately 24,000 customers). Estimated CLV preservation: R28M.
6 Months	Build retailer-facing analytics portal powered by mart_retailer_scorecard. Give retail partners visibility into their own RTS and SLA performance.

This portfolio was built as a demonstration of end-to-end data warehouse engineering capability. All data is synthetic but structurally and statistically representative of real-world last-mile logistics operations. The platform is ready for deployment on Snowflake (primary) or PostgreSQL (open-source alternative) with minimal configuration changes.